

For example, the Prolog facts in such a program might include:

instructor(char, math273)
instructor(patel, ee222)
instructor(grossman, cs301)
enrolled(kevin, math273)
enrolled(~~kevin~~ juana, ee222)
enrolled(juana, cs301)
enrolled(kiko, math273)
enrolled(kiko, cs301)

A new predicate teaches(p, s), representing that professor p teaches student s , can be defined using the Prolog rule

teaches(p, s) :— instructor(p, c), enrolled(s, c)

which means that teaches(p, s) is true if there exists a class c such that professor p is the instructor of class c and student s is enrolled in class c . (Note, that a comma is used to represent a conjunction of predicates in Prolog. Similarly, a semicolon is used to represent a disjunction of predicates.)

Sample Queries

- 1) ? enrolled(kevin, math273) produces the response yes because the fact enrolled(kevin, math273) was provided as input.
- 2) ? enrolled(X , math273) produces the response kevin, kiko.
To produce this response, Prolog determines all possible values of X for which enrolled(X , math273) has been included as a Prolog fact.
- 3) ? teaches(X , Juana). This query returns patel
grossman
Find all the professors who are instructors in classes being taken by Juana.